

Childcare Price Analysis: Final Paper

Summary of Analysis

The National Database of Childcare Prices (NDCP) is the most comprehensive federal source of childcare price data available in the United States. It includes county-level prices by provider type (center-based and family childcare), age group (infants, toddlers, preschoolers, school-age), and county characteristics from 2008 through 2022. Using this dataset and the accompanying technical guide, we focused on full-day center-based infant care (infants aged 0–23 months). Our analysis extracted prices and median household incomes at the county level, aggregated them to the state level, and computed several metrics, including median weekly price, price gap, affordability ratio, and the national trend. We used Python to process the datasets, group counties by state and year, compute mean prices and generate charts.

Findings

The cost of center-based infant care is high and uneven across states. In 2022 the District of Columbia is the most expensive state, with median weekly center-based infant care costing about \$490. Kansas is the least expensive at roughly \$111 per week. The national average in 2022 is roughly \$222 per week, representing a 50% increase since 2008. Family childcare is consistently cheaper than center-based care. Massachusetts exhibits the largest gap, with center-based care being \$130/week more expensive than family care. The affordability burden (the share of median household income consumed by a year of center-based infant care) ranges

from ~30% in Arizona to ~8.7% in Idaho. This indicates that even after accounting for income differences, many families spend a substantial share of their income on infant care.

We also observe a steady upward trend in national average weekly prices: from roughly \$133 in 2008 to \$184 in 2022. Price growth outpaced general inflation, suggesting structural cost pressures in the childcare sector. There is a strong correlation between state incomes and childcare prices: high-income states like DC and Massachusetts pay higher prices but still face significant affordability burdens. Conversely, low-income states like Kansas or Alabama pay lower prices but still devote 10-15% of income to care.

Assumptions

Several assumptions underlie our analysis:

1. **Data completeness:** We assumed the NDCP contains accurate and complete county-level price information for all U.S. states and the District of Columbia. However, some states or counties had missing data; we excluded states without valid price or income information.
2. **Aggregation method:** When summarizing counties to the state level, we used mean values for weekly prices. We assumed this average is representative of the typical county in each state; alternate approaches like population-weighted averages might yield slightly different numbers.
3. **Affordability ratio:** We approximated annual costs as weekly price x 52. This ignores potential seasonal variation or part-year care. Median household income data for 2022 were used to compute affordability, and we assumed incomes were comparable across states and counties.

4. **Focus on infants:** The analysis focused on center-based infant care, as this is the most expensive age group. Toddlers, preschoolers and school-age care were not analyzed in depth.

Items Needing Clarification

- **Missing states:** The NDCP dataset has limited coverage for Puerto Rico and some states in certain years. Clarifying which states or counties have incomplete data would strengthen reliability.
- **Cost adjustments:** Prices were not adjusted for inflation. Future work should standardize prices in real dollars to isolate structural price growth.
- **Provider quality:** The dataset does not record the quality or accreditation of childcare providers. Higher prices might reflect higher quality.
- **Household composition:** Affordability ratios use median household income, not median family income. Clarifying household composition and childcare use patterns would refine affordability estimates.

Story Direction and Message

Our story stresses that childcare costs are both prohibitively high and highly unequal across the country. We highlight the difference between center-based and family care, emphasizing the substantial savings offered by family child-care in many states. We connect price disparities to affordability burdens, demonstrating that some families spend up to a third of their income on infant care. By visualizing trends and state comparisons, we encourage policymakers to target subsidies, tax credits and regulatory reforms to the states and populations

most burdened by childcare costs. Our message is that investing in affordable, high-quality childcare is essential for family financial security and workforce participation.

Target Audience

1. **Policymakers and senior leaders:** The executive brief distils key statistics and recommendations for decision makers who need concise, actionable information.
2. **Analysts and program managers:** The Power BI dashboard allows data professionals to explore county-level patterns, evaluate program impacts and identify hotspots requiring resources.
3. **Community stakeholders and advocacy groups:** The slide deck provides a narrative format to engage parents, educators and community leaders, helping them understand trends and advocate for change.

Mediums Included and Why

- **Executive Brief (PDF):** A single-page document summarizes key metrics, findings and policy recommendations. Policymakers often prefer concise briefs that can be read quickly and shared easily.
- **Interactive Dashboard (Power BI):** Analysts require tools to drill into data by geography, year, provider type and age group. The dashboard uses filters, maps, line and bar charts to help users explore patterns without programming knowledge.
- **Slide Deck (PowerPoint):** Presentations are effective for mixed audiences. The deck provides a narrative flow supplemented with visuals that make abstract statistics concrete.

Design Decisions

We employed a professional yet accessible aesthetic across all mediums. The color palette uses neutral backgrounds with blue and green highlights, reinforcing calm and trust. Charts were kept simple and labelled clearly, horizontal bar charts were used for state comparisons, line charts for trends and box plots for distributions. In the dashboard, filters and legends were placed logically to facilitate exploration. We avoided clutter by balancing text and graphics, leaving white space to guide the reader's eye. Captions and annotations accompany visuals to prevent misinterpretation.

Ethical Considerations

The NDCP data are public and aggregated at the county level, so they do not include personally identifiable information. Nonetheless, our analysis adhered to several ethical principles:

- **Privacy and confidentiality:** We did not disaggregate data to the individual or household level. County-level and state-level aggregates prevent identification of specific providers or families. Where counties or states lacked sufficient data, we excluded them rather than impute values.
- **Transparency:** We clearly state our methods and assumptions, including how we calculated affordability ratios and aggregated data. This transparency helps readers judge the reliability of the findings.
- **Regulatory compliance:** The NDCP is compiled by the U.S. Department of Labor and is intended for public use. We used the data in accordance with its license, cited original sources, and did not misrepresent or monetize the data.

Data Transformations and Filtering

We transformed the NDCP data by aggregating county-level weekly prices and median incomes to the state level and computing price gaps and affordability ratios. These transformations were clearly labelled in the dashboard and paper. We filtered out states with missing price or income data and focused on the 2022 cohort. No hidden filtering was performed.

Legal and Regulatory Guidelines

Childcare data in the U.S. are subject to federal and state licensing regulations that govern provider qualifications, safety and data reporting. However, there are no specific legal restrictions on analyzing publicly released NDCP data beyond citing sources and protecting individual privacy. Our use complies with the open data policy of the Department of Labor.

Data Credibility and Acquisition

The NDCP is produced by the U.S. Department of Labor and is based on market price surveys conducted by states and territories. The data's credibility stems from its governmental authority and methodological transparency. We verified that aggregated results matched published statistics.

Mitigation of Ethical Risks

To mitigate potential ethical issues:

- We contextualized price data by providing affordability ratios, highlighting that high prices often reflect high operating costs rather than provider profiteering.

- We recommended that policymakers use these data to increase subsidies and expand family child-care options, aiming to reduce financial stress on families rather than penalize providers.
- We emphasized limitations and encouraged future research to incorporate inflation adjustments, quality ratings and demographic factors.

Lessons Learned

Through this project we learned that data storytelling is as important as data analysis. Cleaning and aggregating a large dataset like the NDCP requires careful assumptions and documentation. Visualizing complex information across multiple mediums (executive brief, dashboard and slides) helped tailor messages to diverse audiences. We also recognized the importance of ethical transparency: articulating transformations and acknowledging limitations build trust.

If undertaking a similar project again, we would incorporate inflation adjustments to express prices in real dollars, explore county-level patterns in more depth, and examine other age groups and provider types. We might also design interactive tools earlier in the process to guide our analysis and gather feedback sooner. Finally, we would engage with subject-matter experts - for example, early childhood educators and economists - to interpret findings and refine recommendations.

We enjoyed exploring a rich dataset that has direct implications for families and policymakers and translating complex analyses into clear, actionable insights. The creative process of designing visuals and narratives reinforced the power of storytelling to inspire meaningful conversation and potential policy change.